ML Analysis Pipeline (run_all_analyses.py)
7-Tier Scientific Workflow

TIER 1: Independent Analyses

- Demographics Analysis
- Institutional Statistics
- Feature Engineering
- Cancer Type Summary
- Variant Analysis

TIER 2: Data Setup

- 2a: Multi-Tier Groupings (OncoTree hierarchy)
- 2b: Train/Test Split 80% / 20% (stratified by cancer type)

- TRAINING DATA (80%) Used for TIER 3-4
- TEST DATA (20%) Held out until TIER 5

TIER 3: Hyperparameter Optimization

- RandomizedSearchCV (training data only)

  Result: Random Forest AUC = 0.5913

TIER 4: Feature Set Comparison

- Compare Feature Sets (training data only)

  demographic vs technical vs both

first use

TIER 5: Final Model Training

- Train Random Forest on Training Data (80%)
- Evaluate on Test Data (20%) FIRST AND ONLY USE
- Save Models (.joblib files)

TIER 6: SHAP Analysis

- 6a: Primary Feature Values

  Mean |SHAP| rankings
  Beeswarm plots
  Bar charts

- 6b: Feature Interactions

  Pairwise interactions
  Heatmaps

TIER 7: Post-Hoc Statistics

- Age by Actionability

  Mann-Whitney U test
  FDR correction

- SHAP Feature Associations

  Fisher's exact test
  Chi-square test

Output Directory
runs/run_YYYYMMDD_HHMMSS/

tier1_independent_analyses/
tier2_data_setup/
tier3_hyperparameter_optimization/
tier4_feature_comparison/
tier5_final_models/
tier6_shap_analysis/
tier7_posthoc_statistics/